

Vertex Example

The diagram shows the sum of three Feynman diagrams representing the Ward identity for a vertex correction. The first diagram shows a fermion line with a photon loop and a ghost loop, with a dashed line and a wavy line attached to the vertices. The second diagram is similar but with the wavy line and dashed line swapped. The third diagram shows a fermion line with a photon loop and a ghost loop, with a wavy line and a dashed line attached to the vertices. The sum of these three diagrams is equal to zero.

$$\text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} = 0$$